

## Chapter 16: Symmetry

### Worksheet 16: Symmetry

**Learning Objective:** Identify line symmetry and rotational symmetry in figures, determine the order and angle of rotational symmetry, and relate symmetry to natural and artistic patterns.

**Worked Example:**

Find the order of rotational symmetry of a square.

*Solution:* A square looks identical after rotations of  $90^\circ$ ,  $180^\circ$ ,  $270^\circ$ , and  $360^\circ$ . So its order of rotational symmetry is 4.

**Questions (1–20)**

1. How many lines of symmetry does a regular hexagon have?
2. Does the letter “S” have line symmetry?
3. What is the order of rotational symmetry of an equilateral triangle?
4. Define “angle of rotation” for a shape with rotational symmetry.
5. How many lines of symmetry does a (non-equilateral) isosceles triangle have?
6. Find the order of rotational symmetry of a regular pentagon.

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7. What is the angle of rotation for a shape with rotational symmetry of order 6?
  8. Does a (non-rectangular) parallelogram have line symmetry? Does it have rotational symmetry?
  9. A figure has rotational symmetry of order 4. What is its angle of rotation?
  10. Name a capital letter that has both line symmetry and rotational symmetry.
  11. A wheel with 8 equally spaced spokes is rotated. Find its order of rotational symmetry and angle of rotation.
  12. A figure has rotational symmetry of order 5. If one position is at  $0^\circ$ , list all the angles at which it looks the same.
  13. A rangoli design has 6-fold rotational symmetry. If a red dot is placed at one point, how many more identical dots should appear to complete the design?

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14. Compare the order of rotational symmetry of a square and a (non-square) rectangle.
  
  
  
  
  
  
  
  
  
  
  15. A regular polygon has an angle of rotation of  $45^\circ$ . How many sides does it have?
  
  
  
  
  
  
  
  
  
  
  16. A circular clock face has 12 equally spaced numbers. Find the order of rotational symmetry of the numbers' positions, and the angle of rotation.
  
  
  
  
  
  
  
  
  
  
  17. A wallpaper motif has both a vertical line of symmetry and rotational symmetry of order 2. Describe what such a motif might look like, and how you would check both types of symmetry.
  
  
  
  
  
  
  
  
  
  
  18. A regular polygon has 9 lines of symmetry and rotational symmetry of order 9. Name the polygon and find each interior angle.
  
  
  
  
  
  
  
  
  
  
  19. Find an object or design at home with rotational symmetry (a ceiling fan, a flower, a wheel). Estimate its order of rotational symmetry.

20. Look at a rangoli, mandala, or traditional textile design from your region. Describe whether it shows line symmetry, rotational symmetry, or both.

**Reflection Question:** Why do you think many natural objects (flowers, starfish, snowflakes) and traditional art forms show symmetry? Write your own idea.

**Real-Life Application Question:** Fold and cut a paper shape to create a design with line symmetry, OR draw a simple pinwheel shape and check its order of rotational symmetry.